Lab Report: Image classification experiments

1 Introduction

In this report we compare several models on standard benchmarks. We trained each model for 100 epochs with a batch size of 32. The learning rate was set to 0.001 with the Adam optimizer.

2 Results

Figure 1 shows the accuracy during training.

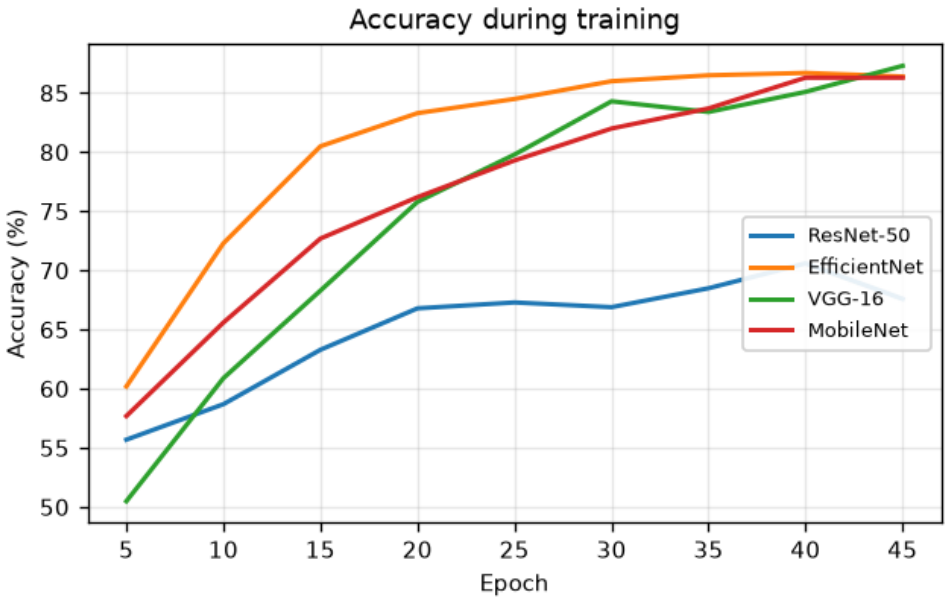


Figure 1: Accuracy during training for the models in this experiment.

As shown in Figure 1, By epoch 45, the accuracy of ResNet-50 stands at 87.8%. Epoch 20 sees MobileNet post a accuracy of 76.2%. VGG-16 attains a accuracy of 42.4% at epoch 5. MobileNet beats ResNet-50 by about 19 points at epoch 45.

Figure 2 shows the f1 score by dataset.

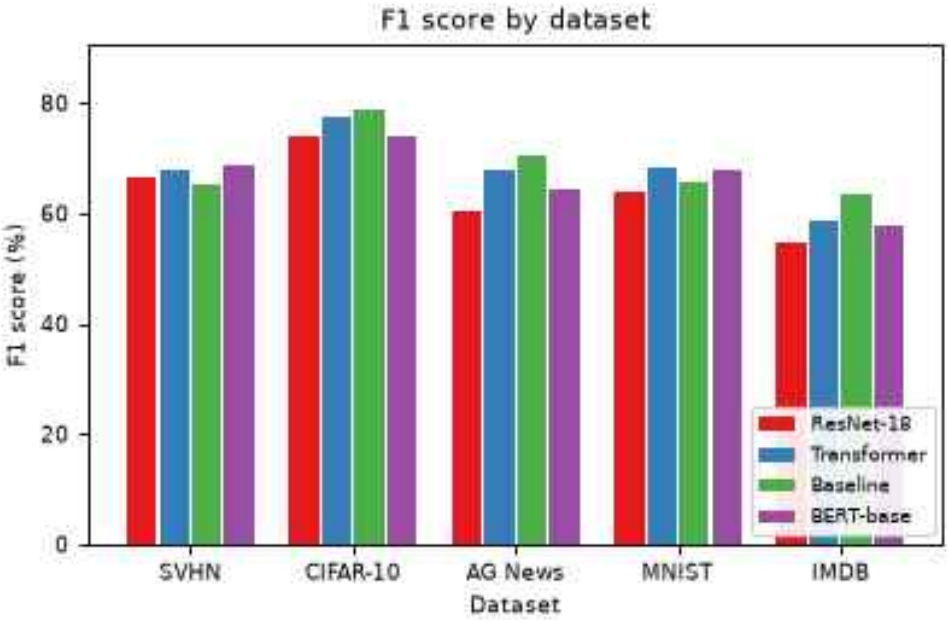


Figure 2: F1 score by dataset for the models in this experiment.

As shown in Figure 2, on MNIST, BERT-base leads ResNet-18 by 4.1 points. On SVHN, Transformer leads

ResNet-18 by 5.9 points.

Figure 3 shows the accuracy of ResNet-18 by dataset.

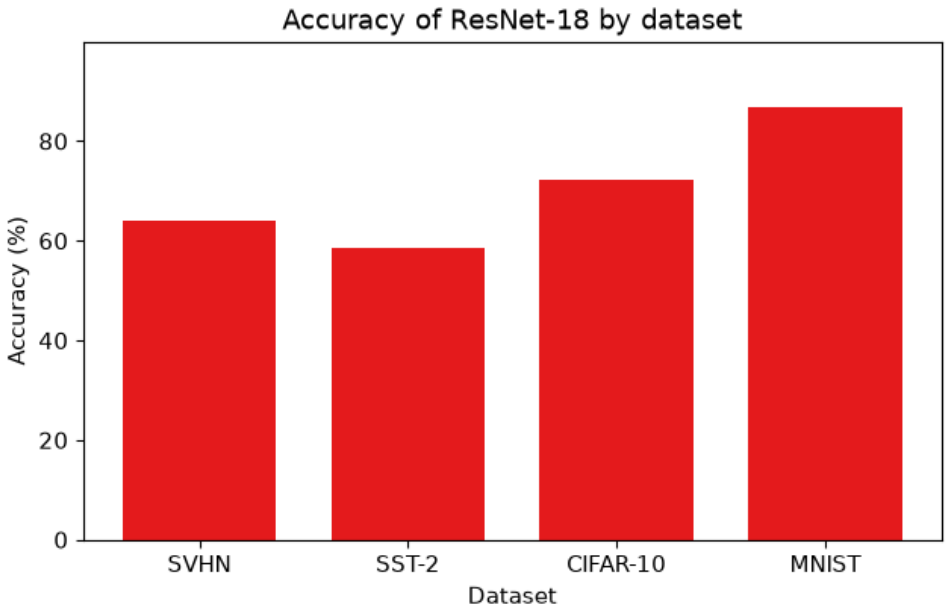


Figure 3: Accuracy of ResNet-18 by dataset for the models in this experiment.

ResNet-18 is 3.1 points higher on CIFAR-10 than on SVHN. As shown in Figure 3, on CIFAR-10, ResNet-18 obtains a accuracy of 77.1%. As shown in Figure 3, ResNet-18 reaches 14.0% accuracy on SVHN.

3 Discussion

On SVHN, XGBoost obtains an accuracy of 53.9%. Overall, the results are in line with what we expected from the literature.